

# Macroeconomics III

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...also based on  
Chapter 3 of your  
textbook

“CAPITAL  
ACCUMULATION AND  
GROWTH: THE BASIC  
SOLOW MODEL”

## THE COMPLETE MODEL (with book equation numbers below)

$$Y_t = BK_t^\alpha L_t^{1-\alpha} \quad (14)$$

$$r_t = \alpha B \left( \frac{K_t}{L_t} \right)^{\alpha-1} \quad (15)$$

$$w_t = (1-\alpha) B \left( \frac{K_t}{L_t} \right)^\alpha \quad (16)$$

$$S_t = sY_t \quad (17)$$

$$K_{t+1} - K_t = S_t - \delta K_t \quad (18)$$

$$L_{t+1} = (1+n)L_t, \quad (19)$$

- Parameters:  $B, \alpha, s, \delta$  and  $n$ . N.b. No subscript  $t$  on  $B$ : "Basic" Solow model.
- Endogenous variables:  $(Y_t), (K_t), (L_t), (r_t), (w_t)$  and  $(S_t)$  of which  $(K_t)$  and  $(L_t)$  are state variables:

## GOVERNMENT SECTOR

- $K_{t+1} - K_t = S_t - \delta K_t$  follows from the identity/definition  $K_{t+1} - K_t = I_t - \delta K_t$  and the national accounting identity  $Y_t = C_t + I_t \Leftrightarrow Y_t - C_t = I_t \Leftrightarrow S_t = I_t$
- Same construction with a public sector:

$$\begin{aligned} Y_t &= C_t^p + C_t^g + I_t^p + I_t^g \Leftrightarrow \\ &\underbrace{Y_t - T_t - C_t^p}_{S_t^p} + \underbrace{T_t - C_t^g}_{S_t^g} = I_t^p + I_t^g \end{aligned}$$

- Insert into the identity  $K_{t+1} - K_t = I_t^p + I_t^g - \delta K_t$  to get

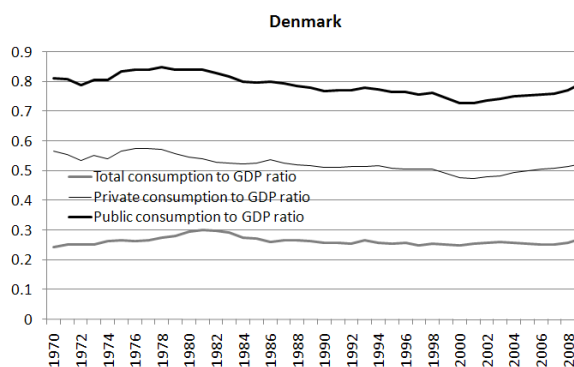
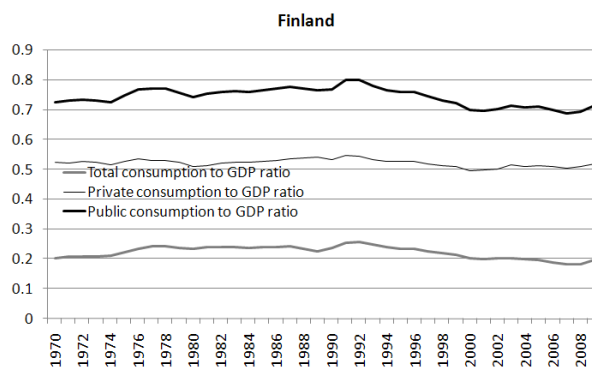
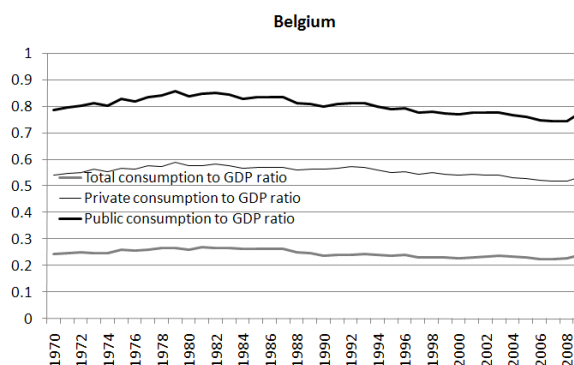
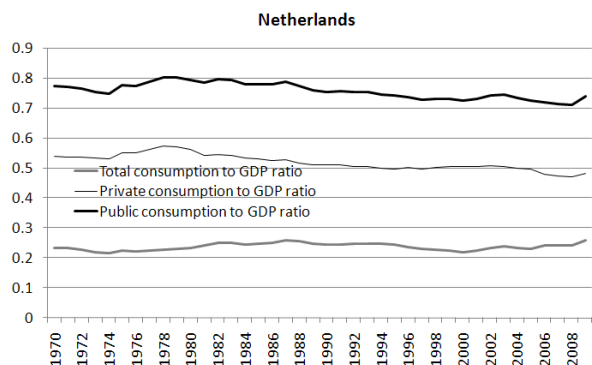
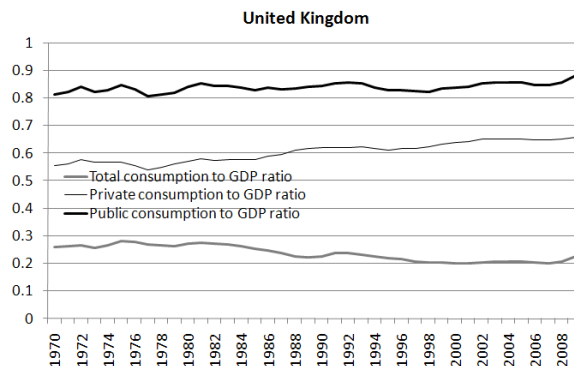
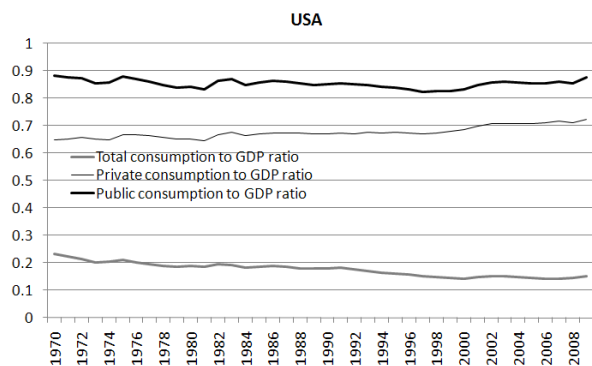
$$K_{t+1} - K_t = \overbrace{S_t^p + S_t^g}^{S_t} - \delta K_t$$

- We can use the model as it stands. We just have to reinterpret  $K_t$  as the sum of private and government capital stock and  $S_t$  as the sum of private and public savings. Similarly the equation  $S_t = sY_t$  should be reinterpreted as:

$$S_t = S_t^p + S_t^g = sY_t \Leftrightarrow C_t^p + C_t^g = (1-s)Y_t$$

- The essential assumption underlying the Solow model interpreted to include a government is that **the sum of private and public consumption as a fraction of GDP is a constant,  $1-s$** .
- This seems plausible empirically. And  $s \approx 0.2$  seems plausible for many Western countries:

# The consumption share of GDP in several Western countries



# THE BASIC SOLOW MODEL, SHORT VERSION

$$Y_t = BK_t^\alpha L_t^{1-\alpha}$$
$$K_{t+1} - K_t = sY_t - \delta K_t$$
$$L_{t+1} = (1+n)L_t$$

- In the previous part we saw that the Solow model leads to the **transition equation**:

$$k_{t+1} = \frac{1}{1+n} (sy_t + (1-\delta)k_t)$$

- and to the **Solow equation**:

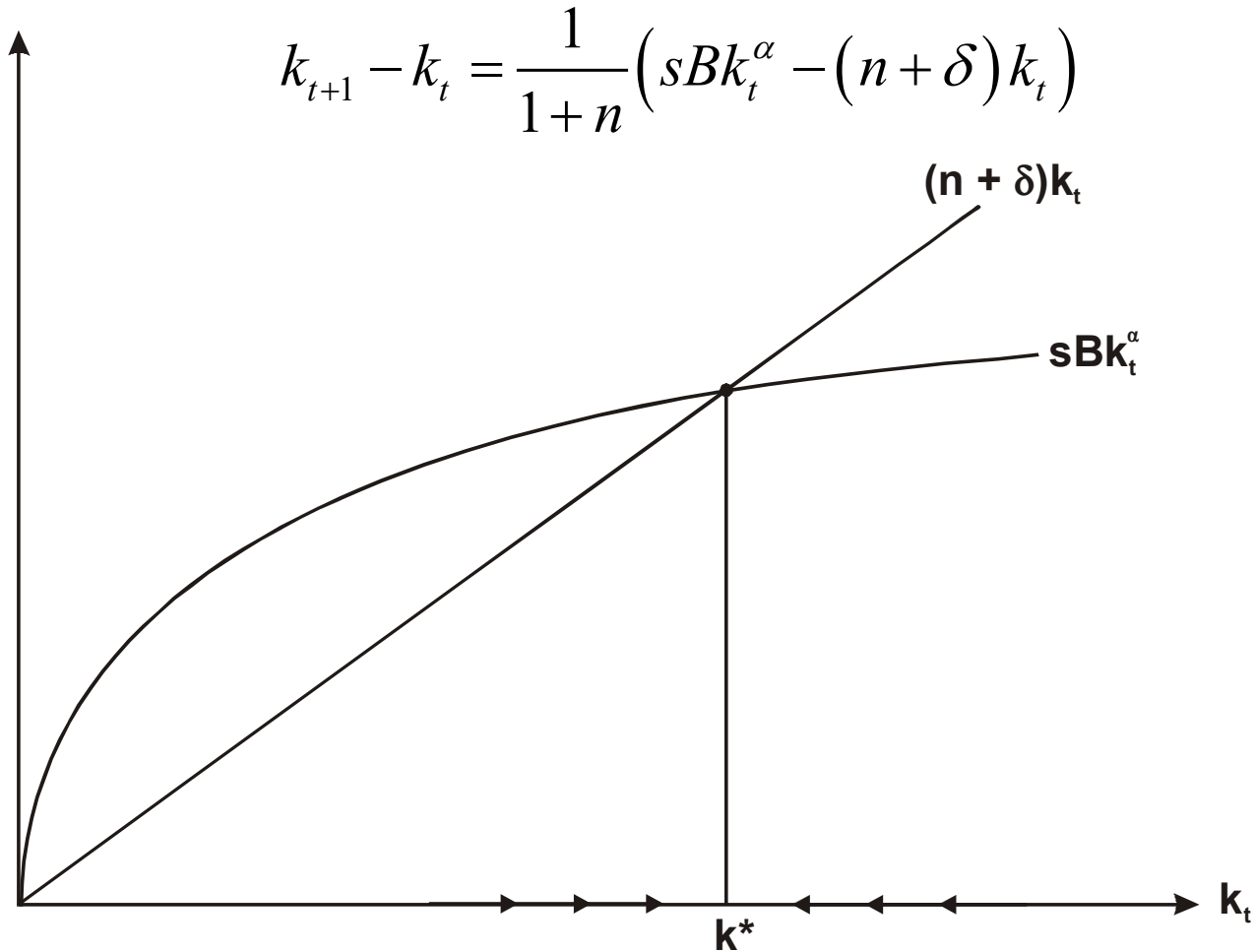
$$k_{t+1} - k_t = \frac{1}{1+n} \left( \underbrace{sy_t}_{\text{savings per capita}} - \underbrace{(n+\delta)k_t}_{\text{Replacement investment to compensate for depreciation and growth of labour force}} \right)$$

"technical term"  
appearing because  
of discrete time

savings per  
capita =  $sy_t$

Replacement investment to  
compensate for depreciation  
and growth of labour force

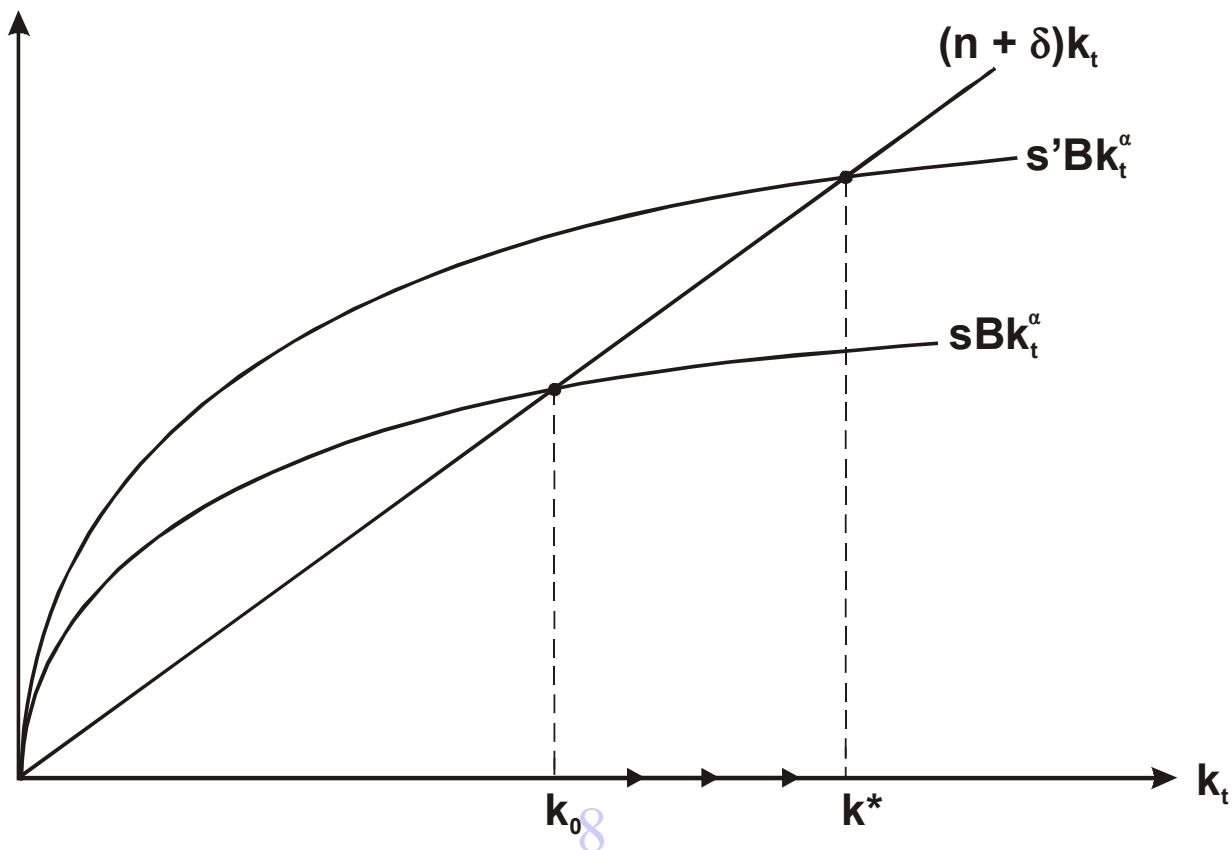
## The Solow diagram (repetition)



Why does growth in  $k_t$  and  $y_t$  have to stop?  
Diminishing returns!

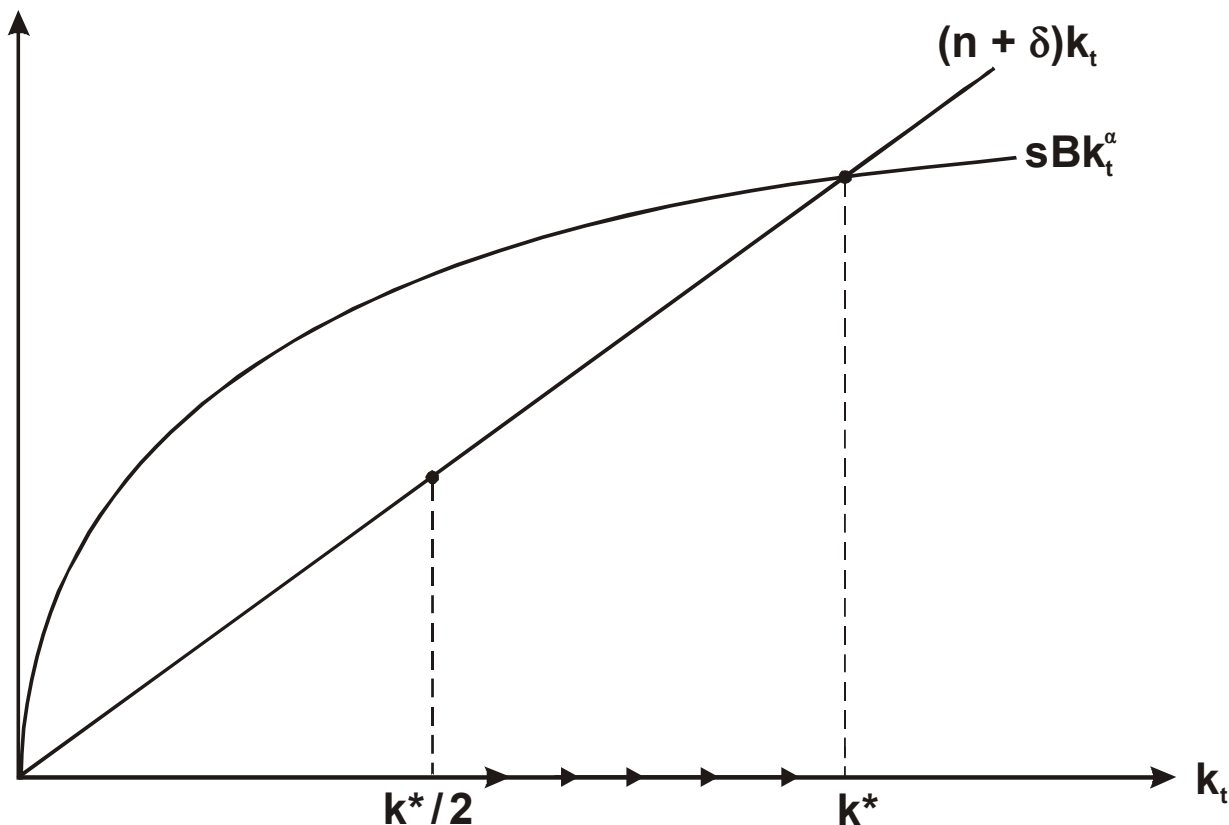
# COMPARATIVE ANALYSIS IN THE SOLOW DIAGRAM

1. The economy is initially in steady state with parameters  $B, \alpha, s, \delta$  and  $n$ . What happens if the savings rate increases permanently from  $s$  to a new and higher level,  $s'$ ?





2. The economy is initially in steady state. No parameters change, but an exogenous event, e.g., a war or natural disaster, reduces the capital stock to half size "overnight". How is this analysed in the Solow diagram?



# STEADY STATE

- Last time we saw that the Solow model implies convergence to a unique steady state. From the Solow equation

$$k_{t+1} - k_t = \frac{1}{1+n} \left( sBk_t^\alpha - (n + \delta)k_t \right)$$

one easily computes

$$k^* = B^{\frac{1}{1-\alpha}} \left( \frac{s}{n + \delta} \right)^{\frac{1}{1-\alpha}}$$

and then

$$y^* = B(k^*)^\alpha = B^{\frac{1}{1-\alpha}} \left( \frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

$$r^* = \alpha B(k^*)^{\alpha-1} = \alpha \left( \frac{s}{n + \delta} \right)^{-1}$$

$$w^* = (1 - \alpha) B^{\frac{1}{1-\alpha}} \left( \frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}} = (1 - \alpha) y^*$$

## Some main lessons (repeated from previous note)

$$\ln y^* = \frac{1}{1-\alpha} B + \frac{\alpha}{1-\alpha} [\ln s - \ln(n + \delta)].$$

- The elasticity of  $y^*$  wrt.  $s$  is  $\frac{\alpha}{1-\alpha} \approx \frac{1}{2}$ :  
an increase in the savings rate of 10%, e.g., from 20 to 22%, gives a long-run increase in income per worker of around 5% according to the basic Solow model!
- The elasticity of  $y^*$  wrt.  $B$  is  $\frac{1}{1-\alpha} \approx \frac{3}{2} > 1$  ! Note that the effect is stronger than one-to-one due to capital accumulation.
- Another steady state prediction concerns the real interest rate...

## THE "NATURAL" INTEREST RATE

$$r^* = \alpha \left( \frac{s}{n + \delta} \right)^{-1} \Rightarrow \rho^* = \alpha \left( \frac{s}{n + \delta} \right)^{-1} - \delta$$

- The real rate of interest is determined by productivity and thrift in the long run (Knut Wicksell):
  - Higher  $\alpha \Rightarrow$  capital is more productive  $\Rightarrow$  demand of capital per worker increases (ceteris paribus)  $\Rightarrow$  higher equilibrium interest rate (the price of capital).
  - Higher  $s / (n + \delta) \Rightarrow$  supply of capital per worker increases  $\Rightarrow$  the equilibrium interest rate decreases.
- Reasonable parameter values on an annual basis,  $\alpha = 1/3$ ,  $s = 0.22$ ,  $n = 0.005$ ,  $\delta = 0.05$ , imply  $r^* = 8.3\%$  and  $\rho^* = 3.3\%$ . This value for  $\rho^*$  is very close to empirical observations of the real interest rate!

# STRUCTURAL POLICY

## 1. Crowding out:

- Consider a permanent fall in  $s$  caused by a permanent increase in government consumption as a percentage of GDP.
- What happens **on impact**?  $Y_t$  is unaffected and still grows at the rate of  $n$ , and  $y_t$  is unaffected. But savings decrease and consumption increases. There is **full crowding out**.
- What happens **in the longer run**? During a transitional period  $Y_t$  grows more slowly than at the rate of  $n$  and  $y_t$  falls down to a new lower steady state level. There is **more than full crowding out**. And the real interest rate increases.
- The government cannot increase GDP by raising government expenditure in the long run. How about the short run? (Keynes...)

## 2. Motives for tax-financed public services from a long-run perspective

- Public investments (that would not be made by private agents)

For government consumption:

- Public (non-rival and possibly non-excludable) goods
- Public consumption, e.g., on education and health care, replacing private consumption, which means that  $s$  is not affected
- Distributive reasons
- Externalities (education)
- General productivity effects of public consumption, e.g., judicial system, health care, etc.

### 3. Incentive policies:

- Policies that do not affect model parameters directly through government expenditure/revenue, but indirectly through the way they affect private behaviour. We cannot analyse incentive policies explicitly because private behaviour has not been derived from optimization.
- Golden rule:

$$y^* = B^{\frac{1}{1-\alpha}} \left( \frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}} \Rightarrow$$
$$c^* = B^{\frac{1}{1-\alpha}} (1-s) \left( \frac{s}{n + \delta} \right)^{\frac{\alpha}{1-\alpha}}$$

The  $s$  that maximizes  $c^*$ , which is  $s^{**} = \alpha$ , is called the golden rule savings rate.

- The model suggests structural policies that
  - promote technology
  - encourage savings (assuming that  $s$  is considerably below  $s^{**}$ ): institutions and incentive
  - reduce  $n$

# GROWTH IN THE BASIC SOLOW MODEL

- The long run prediction of the Solow model is its steady state. What is the growth rate of GDP per capita in steady state?
- Zero! Not in accordance with stylized facts.
- What is the growth rate of  $Y_t$  in steady state? Since  $Y_t / L_t = y_t$  and  $y_t$  is constant,  $Y_t$  and  $L_t$  must grow at the same rate,  $n$ . GDP grows, but only at the same rate as the labour force. Why is that?
- Assume that the economy initially is below steady state, implying that  $k_t < k^*$ . Then  $sBk_t^\alpha > (n + \delta)k_t$ , implying that capital per worker increases. But, once again, because of **diminishing returns** the growth in  $k_t$  and  $y_t$  will ultimately cease.
- However, there is transitory growth. How long-lasting is that?



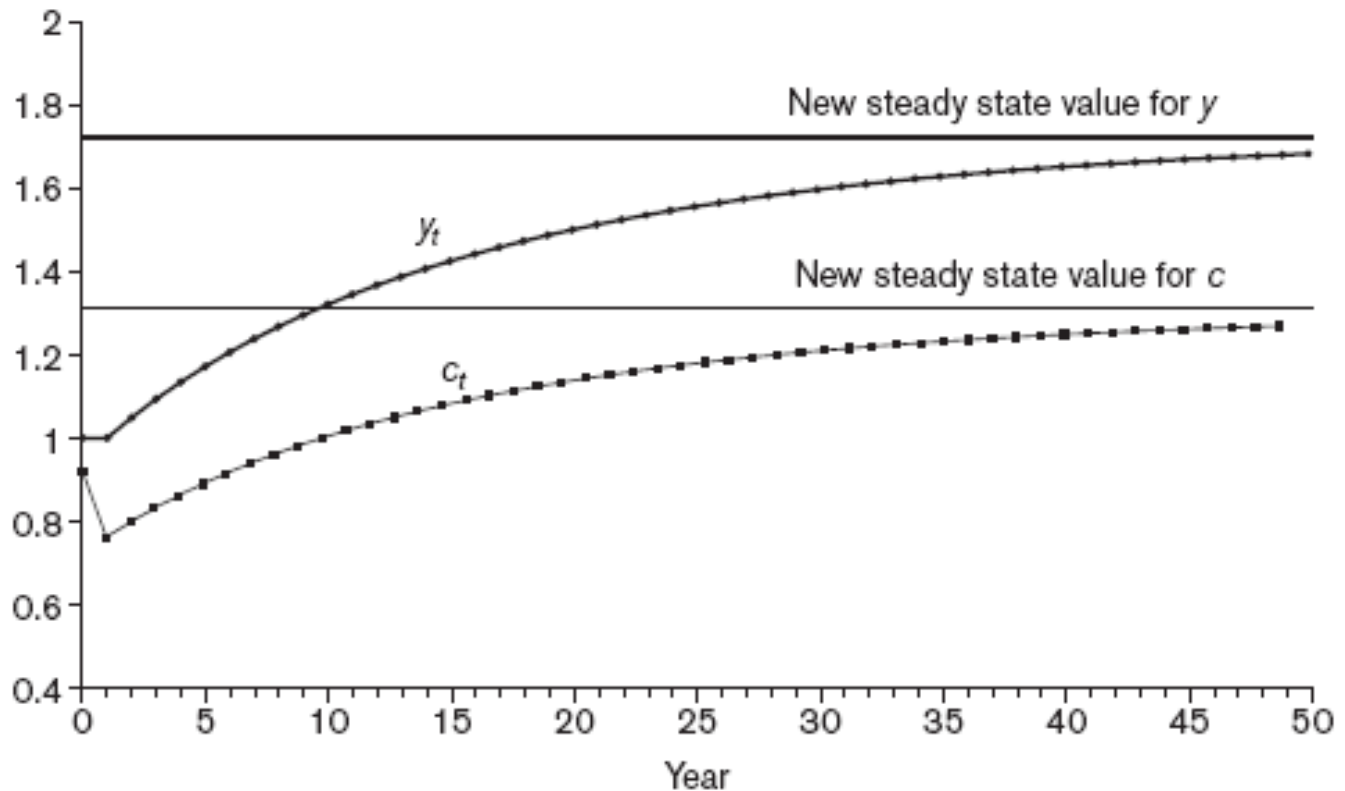
# SIMULATION

- Initially we are in steady state with the following parameter values:  $B = 1$ ,  $\alpha = 1/3$ ,  $\delta = 0.05$ ,  $n = 0.03$ ,  $s = 0.08$  representing a developing country. This implies that  $k^* = y^* = 1$ .
- With effect first time in period 1, the savings rate increases permanently to  $s' = 0.22$  corresponding to the savings rate of a typical Western economy, implying  $k^{*'} = 5.20$  and  $y^{*'} = 1.73$ .
- Starting with  $k_0 = 1$  in period zero and one we now simulate

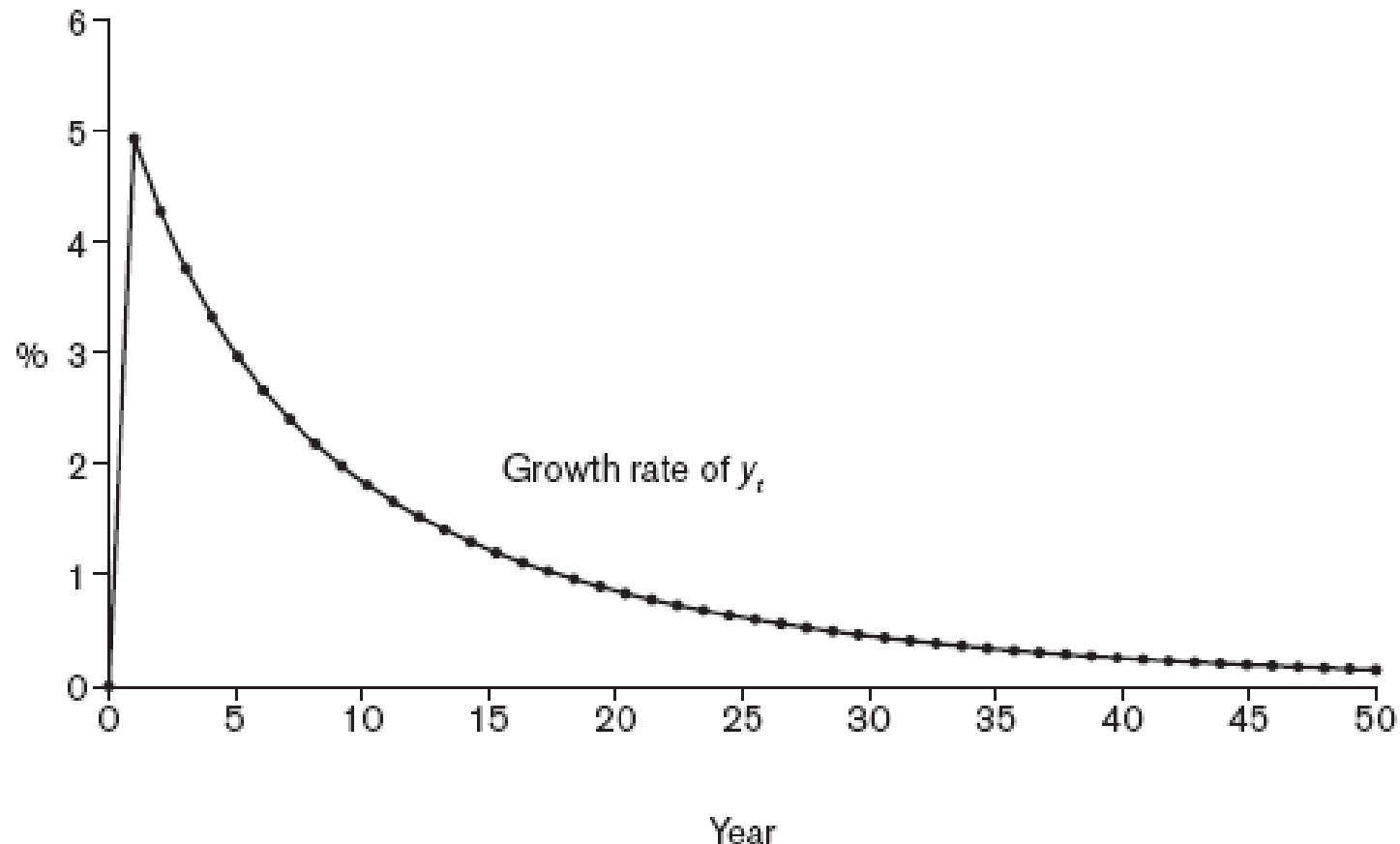
$$k_{t+1} = \frac{1}{1+n} \left( s' B k_t^\alpha + (1-\delta) k_t \right)$$

over  $t = 2, 3, \dots$ . We also calculate  $y_t = B k_t^\alpha$ ,  $c_t = (1-s') y_t$  and  $g_t^y = \ln y_t - \ln y_{t-1}$  etc. for  $t = 1, 2, 3, \dots$  and draw the evolutions of these variables in the following figures.

The evolution of  $y_t$ ,  $c_t$  and  $g_t^y$  after the increase in  $s$



The evolution of  $y_t$ ,  $c_t$  and  $g_t^y$  after the increase in  $s$  (continued)



The figures show that transitory growth is relatively long-lasting.

- In the Solow model, the transition towards steady state is at least as important as the steady state itself. And during this transition there is growth in  $k_t$  and  $y_t$ . Hence, **the basic Solow model is a growth model!**
- It is easy to find the growth rate of  $k_t$  :

$$k_{t+1} - k_t = \frac{1}{1+n} \left( sBk_t^\alpha - (n + \delta)k_t \right) \Rightarrow$$

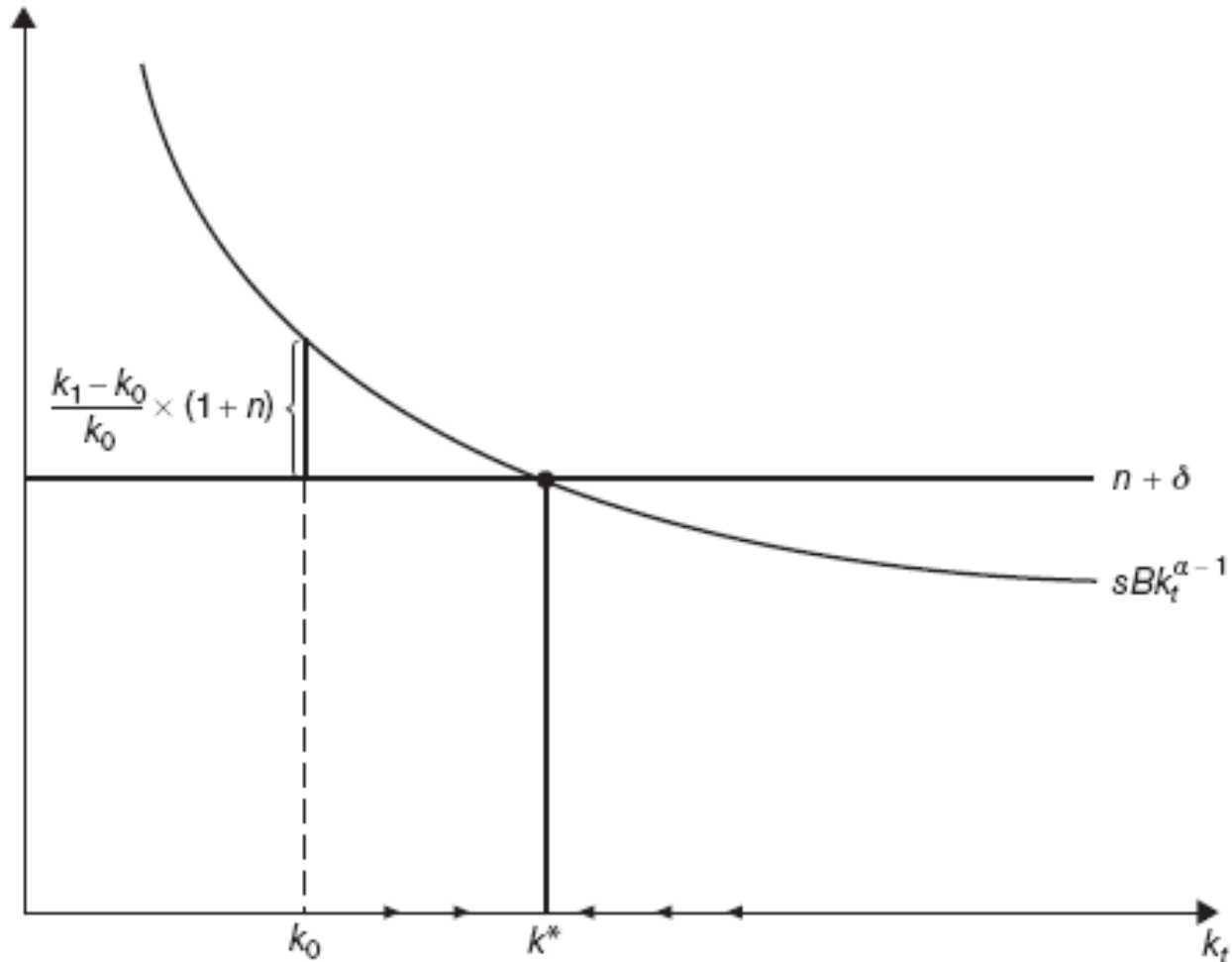
$$\frac{k_{t+1} - k_t}{k_t} = \frac{1}{1+n} \left( sBk_t^{\alpha-1} - (n + \delta) \right)$$

This is called the **modified Solow equation**.

- The growth rate of  $y_t$  follows from  $g_t^y = \alpha g_t^k$  .

## The modified Solow diagram

$$\frac{k_{t+1} - k_t}{k_t} = \frac{1}{1+n} \left( sBk_t^{\alpha-1} - (n + \delta) \right)$$



- Growth in GDP per worker is higher the further below steady state the economy is. This is in accordance with **conditional convergence**.
- A permanent increase in  $s$  gives a jump upwards in the growth rate of GDP per worker.

# CONCLUSIONS BASED ON THE BASIC SOLOW MODEL

- What can a (poor) country do to create a transitory growth in GDP per worker resulting in a permanently higher level of income and consumption per worker? The basic Solow model provides the following answers:
  - Increase the savings rate
  - Reduce the growth rate of the labour force
  - Reduce the rate of depreciation, i.e., invest better
  - Improve the level of technology
- How useful are these recommendations?